

SYSTEM VERIFICATION AND VALIDATION PLAN

CSI-100 Cell Sorting Instrument (IVD)

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Purpose

This plan defines how the CSI-100 system is verified against the system requirements in SRS-CSI-001 (design outputs meet design inputs) and validated against user needs and intended use (the finished device meets user needs), per 21 CFR 820 and ISO 13485:2016 clauses 7.3.6 and 7.3.7.

Document Control

Rev	Date	Description of change	Author
A	Sep 2026	Initial release. Based on SRS-CSI-001 Rev D (verification baseline) and user needs UN-01 to UN-16.	A. Lodhi

Approvals

Function	Role	Signature / Date
Systems Engineering	Author, V&V Plan Owner	
V&V Engineering	Verification Lead	
Clinical / Medical Affairs	Validation Lead	
Human Factors	Usability Validation Lead	
Software Engineering	Software Verification Owner	
Quality Engineering	Design Quality, Risk Management	
Regulatory Affairs	Submission Strategy	
Independent Reviewer	Design review independent of the design	

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1. Introduction

1.1 Purpose

This plan defines the strategy, methods, acceptance criteria, responsibilities and schedule for design verification and design validation of the CSI-100 Cell Sorting Instrument, as required by 21 CFR 820 (QMSR) and ISO 13485:2016 clauses 7.3.6 and 7.3.7.

Design verification confirms that design outputs meet design inputs: each system requirement in SRS-CSI-001 is shown to be met by objective evidence. **Design validation** confirms that the finished device meets user needs and its intended use, under actual or simulated use conditions, with production-equivalent units and intended users. The two are planned, executed and reported separately in this document (Part 1 and Part 2).

1.2 Scope

This plan works at the system level. Verification is planned against the system requirements (SYS-nnn) in SRS-CSI-001 Rev D; validation is planned against the user needs (UN-nn) and intended use in the same document. Subsystem and software verification are carried out under their own plans; this plan does not repeat their detail, but it cites their results where they provide evidence for a system requirement or a risk control (for example SRS-SW-CSI-010, STS-CSI-010 and HA-CSI-001).

In scope: the sorter, fluidics supply module, workstation and software release 3.0, accessory kit and the aerosol evacuation, plate deposition and temperature control options, in all configurations of CFG-CSI-001. Out of scope: assay-specific clinical performance, which is validated under each assay's own plan.

1.3 Verification and validation compared

	Design verification (Part 1)	Design validation (Part 2)
Question	Did we build the device right?	Did we build the right device?
Compared against	System requirements SYS-nnn (design inputs)	User needs UN-nn and intended use
Test articles	Design verification units DV-01 to DV-05 (pilot build)	Production-equivalent units VAL-U1 to VAL-U4 (validation build)
Conditions	Controlled laboratory, reference materials, worst-case settings	Actual or simulated use at clinical sites, clinical specimens
Performed by	V&V engineers	Intended users (clinical laboratory scientists), observed by Clinical Affairs and Human Factors
Protocols	DVP-CSI-nnn	VAL-CSI-nn (in DVAL-CSI-001)
Reports	PVR-CSI-001, SVR-CSI-001, VVR-CSI-001 Part 1	DVAL-CSI-001 report, human factors report, VVR-CSI-001 Part 2
Entry criteria	Design frozen at DIR-3; protocols approved	Relevant verification complete; production-equivalent units; final labeling

2. V&V Strategy

2.1 V-model

Figure 1 shows how each level of the design is checked. The shaded boxes are covered by this plan; the dashed boxes are carried out under other plans and referenced here.

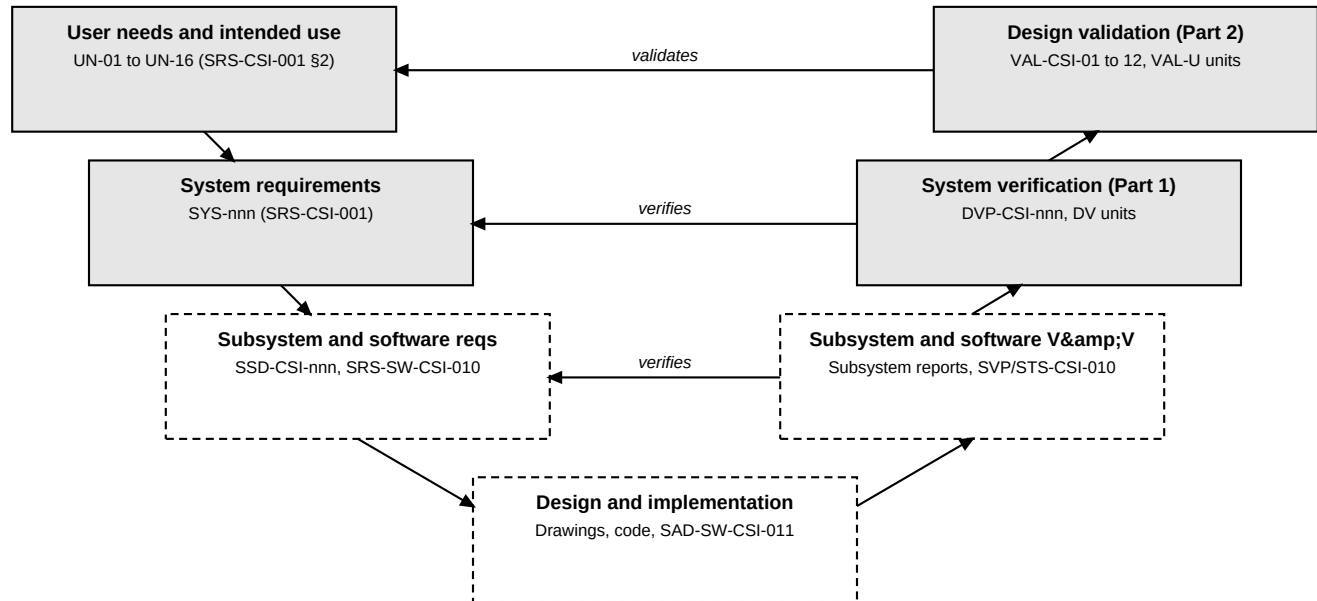


Figure 1. V-model for the CSI-100. Shaded: this plan. Dashed: other plans, referenced.

2.2 Separation of verification and validation

Conflating verification and validation is a common audit finding. This plan keeps them apart with the following rules:

- Different inputs: verification is traced to SYS requirements; validation is traced to UN user needs. Neither trace table uses the other's identifiers (Section 14).
- Different test articles: verification uses DV units from the pilot build; validation uses VAL-U units from the validation (PQ) build defined in DTP-CSI-001, built with released production documentation.
- Different people: verification is executed by V&V engineers; validation is performed by intended users at clinical sites, not by the design team.
- Different protocols and reports: DVP protocols report to PVR-CSI-001 and SVR-CSI-001; VAL protocols report to DVAL-CSI-001.
- No credit transfer: a verification test result is never used as validation evidence, and a validation result is never used to close a system requirement.
- Sequence: a validation study starts only when the system requirements it depends on are verified (Section 11.3).

2.3 Risk-based approach

Test rigor and sample size depend on the risk associated with the requirement. Requirements that implement risk controls in HA-CSI-001 are verified for both implementation and effectiveness (Section 8), and use the higher confidence and reliability levels in Section 5.2.

3. Organization and Responsibilities

Activity	Systems Eng	V&V Eng	Software	Clinical Affairs	Human Factors	Quality	Regulatory
V&V plan and trace matrix	A/R	C	C	C	C	C	C
Verification protocols and execution	A	R	C	I	I	C	I
Software verification (SVP-CSI-010)	C	C	A/R	I	I	C	I
External safety, EMC, laser testing	A	R	I	I	I	C	C
Risk control effectiveness verification	R	R	R	I	C	A	I
Validation protocols and execution	C	I	C	A/R	R	C	C
Summative usability evaluation	C	I	C	C	A/R	C	C
Anomaly review board	R	R	R	C	C	A	I
V&V summary report	A/R	C	C	C	C	R	C
Independent review of V&V results	I	I	I	I	I	A	C

R = Responsible, A = Accountable, C = Consulted, I = Informed

4. Test Articles, Equipment and Materials

4.1 Test articles

ID	Build	Configuration	Used for
DV-01	Pilot (T1)	2-laser (488, 633)	Verification; minimum build
DV-02	Pilot (T1)	6-laser (488/445, 561, 633, 405/375)	Verification; maximum build; safety, EMC, laser testing
DV-03	Pilot (T1)	4-laser (488, 561, 633, 405)	Verification; most common build; transport testing
DV-04	Pilot (T1)	4-laser with plate deposition, aerosol evacuation, temperature control	Option verification; containment
DV-05	Pilot (T1)	5-laser, 230 VAC	Voltage and EMC worst case; transport testing
LT-01 to LT-03	Pilot (T1)	4-laser	Reliability life testing
VAL-U1 to VAL-U4	Validation (T2, PQ lots)	Two 4-laser, one 6-laser, one 2-laser; all options at one site	Design validation only

All test articles run software release candidate 3.0 (or later, with regression per Section 15.3). Hardware and software configuration of each article is recorded before each protocol and under configuration control. Configurations not tested directly are covered by bracketing (minimum, maximum and most common builds), with the rationale documented in each protocol.

4.2 Equipment and reference materials

Item	Use	Control
Calibrated power meter, pressure gauge, balance, thermometer, sound level meter, safety tester	Physical and safety measurements	Calibrated, traceable; calibration valid during use
Environmental chamber	Temperature and humidity testing	Calibrated; mapped
Chicken erythrocyte nuclei (PI and Hoechst stained)	Resolution and linearity	Qualified lots
Alignment, QC, multi-peak and 0.5 µm beads	Sensitivity, detection, QC	Qualified lots, within expiry
Mixed fluorescent bead populations; drop-delay beads	Sort purity and yield; drop delay	Qualified lots
Fluorescent microsphere aerosol challenge kit	Containment	Method validated (TMV-CSI-001)
Cell lines and donor PBMCs	Viability after sort	Characterized; consented donors
Reference flow cytometer (legally marketed IVD)	Comparator for validation method comparison	Maintained and QC'd per manufacturer

PART 1. DESIGN VERIFICATION

Design verification confirms that the CSI-100 design outputs meet every system requirement in SRS-CSI-001 Rev D. Each requirement is verified by the method stated in the SRS, under a DVP protocol, on DV units.

5. Verification Methods and Sample Size

5.1 Methods

Method	Code	Definition	Evidence
Test	T	Quantitative measurement against a stated limit under defined conditions	Recorded data, statistical analysis
Inspection	I	Examination of the device, labeling or documentation against the requirement	Inspection record with photographs or document references
Analysis	A	Calculation, modeling or review of design data	Analysis report, reviewed and approved
Demonstration	D	Operation of a function to show it works as required, without detailed measurement	Executed checklist with pass/fail per step

5.2 Sample size and statistical rationale

Requirement risk	Attribute data (pass/fail)	Variable data	Instruments
High: implements a risk control for a hazard of severity 4 or 5	95% confidence / 95% reliability: 59 trials, 0 failures	95% confidence / 99% coverage tolerance interval within limit	3 DV units
Medium: performance claim or risk control for severity 3	95% / 90%: 29 trials, 0 failures	95% / 95% tolerance interval within limit	3 DV units
Low: other requirements	Minimum 3 trials, 0 failures	All results within limit	1 DV unit (worst-case build)
Inspection, analysis, demonstration	One execution per applicable configuration	—	As applicable

Each protocol states the risk level of each requirement, the resulting sample size, the statistical method and the treatment of outliers, before execution. Results are not re-analyzed with a different method after the data are seen.

6. Verification Protocols

Verification is organized in 22 protocols, one or more per SRS section. Protocols are approved before execution and report to the verification reports listed in Section 16.

Protocol	Title	SRS section	Methods	Test articles and samples	Location / owner
DVP-CSI-301	Physical, electrical and environmental	3.1 to 3.4	T, I, A	DV-01 to DV-03; environmental chamber	V&V Lab; external lab for environmental
DVP-CSI-401	Fluorescence resolution and linearity	4.1, 4.2	T	DV-01 to DV-03 × 3 days × 3 runs	V&V Lab
DVP-CSI-402	Scatter sensitivity and detection	4.3, 4.5	T, I, A	DV-01 to DV-03	V&V Lab
DVP-CSI-403	Excitation (lasers, beam geometry, warm-up)	4.4	T, I, D	DV-01 to DV-05, every installed laser	V&V Lab
DVP-CSI-404	Acquisition rate and sample flow	4.6	T	DV-01 to DV-03 × 3 runs	V&V Lab
DVP-CSI-405	Sort performance (purity, yield, viability, nozzles)	4.7	T, I, D	DV-01 to DV-03 × 10 sorts per nozzle	V&V Lab
DVP-CSI-406	Carryover	4.8	T, D	DV-01 to DV-03 × 20 high/low sequences	V&V Lab
DVP-CSI-407	Sort collection and plate deposition	4.9	T, D	DV-04 (plate unit) × 10 plates per format	V&V Lab
DVP-CSI-408	Fluidics and temperature control	4.10, 4.11	T, I, D	DV-01 to DV-03	V&V Lab
DVP-CSI-501	Electrical safety and EMC	5.1	T	DV-02, DV-05 (worst-case build)	Accredited external lab; report SVR-CSI-001
DVP-CSI-502	Laser safety	5.2	T, I	DV-02 (6-laser build)	Accredited external lab; report SVR-CSI-001
DVP-CSI-503	Interlocks, HV, e-stop, clog and breakoff response	5.3 to 5.5	T, I	DV-01 to DV-03 × 59 trials per response	V&V Lab
DVP-CSI-504	Biohazard containment and decontamination	5.6	T, I	DV-04 with aerosol option × 3 runs × 3 conditions	V&V Lab, biosafety level 2 room
DVP-CSI-505	Mechanical and pressure safety	5.7	T, A	DV-01; tank design analysis	V&V Lab; external lab
DVP-CSI-601	Functional requirements	6	T, D	DV-01 to DV-03; STS-CSI-010 results	V&V Lab; Software V&V
DVP-CSI-701	Software, data integrity and cybersecurity	7	T, I	Release candidate 3.0 on DV-01; STS-CSI-010; penetration test	Software V&V; external security tester
DVP-CSI-801	Interfaces	8	T, I, D	DV-01 to DV-04 with options	V&V Lab
DVP-CSI-901	Usability requirements (design verification)	9	T, I, D	DV-01; internal trained participants	Human Factors
DVP-CSI-1001	Reliability, maintainability and serviceability	10	A, T, D	Life test units LT-01 to LT-03; reliability prediction	Reliability Engineering; Service
DVP-CSI-1101	Labeling and documentation	11	I	Final labeling set	Regulatory, Quality

Protocol	Title	SRS section	Methods	Test articles and samples	Location / owner
DVP-CSI-1201	Packaging, transport and installation	12	T, I, D	DV-03 and DV-05 in shipping configuration	Accredited transport lab; Service
DVP-CSI-1301	Regulatory and standards compliance	13	A, I	Compliance checklists, DHF review	Regulatory, Quality

7. Key Performance Acceptance Criteria

For most requirements the acceptance criterion is the requirement statement itself. The requirements below have a statistical acceptance criterion defined here so that it is fixed before testing.

Requirement	Test design	Acceptance criterion
SYS-401 PI CV < 3.0%	3 units × 3 days × 3 runs (27 runs)	Every run < 3.0%; 95/95 upper tolerance limit < 3.0%
SYS-402 Hoechst CV < 3.5%	3 units × 3 days × 3 runs, 405 and 375 nm	Every run < 3.5%; 95/95 upper tolerance limit < 3.5%
SYS-411 Linearity 1.95 to 2.05	27 runs per excitation	Every run within 1.95 to 2.05
SYS-421 0.5 µm beads resolved	3 units × 3 runs	Bead population separated from noise with valley below 10% of peak
SYS-451 70,000 events/s	3 units × 3 runs × 10 min	No lost events; electronic abort count within specification
SYS-461 Purity > 98%, yield > 80%	3 units × 10 four-way bead sorts at 70 PSI, 90 kHz, 25,000 events/s	Lower 95% confidence bound of purity > 98%; of yield > 80% of Poisson
SYS-466 Viability	Cell line and PBMC sorts at each default pressure	Viability ≥ 85% after sort and proliferation over 3 days
SYS-471 Carryover ≤ 0.5%	20 high-concentration then blank sequences per unit	95% upper confidence bound of carryover ≤ 0.5%
SYS-483 Single-cell deposition	10 plates per format (96, 384)	≥ 95% of target wells with exactly one bead, lower 95% bound
SYS-532, 543 E-stop and clog response	59 trials per response per unit	All actions complete within 2 s in every trial
SYS-551 Containment	3 runs × 3 conditions (normal sort, simulated clog, door cycle)	No challenge particles detected outside the sort chamber
SYS-331 Noise ≤ 62 dBA	3 units, sorting at 70 PSI	Every reading ≤ 62 dBA

8. Risk Control Verification

ISO 14971 requires verification that each risk control is implemented and effective. Table 1 lists the hazards from HA-CSI-001 that are controlled by the design, the system requirement that specifies the control, the software requirement where software contributes (SRS-SW-CSI-010), and the verification evidence for implementation and effectiveness. Residual risk is evaluated in RMR-CSI-001.

Hazard (HA-CSI-001)	System requirements	Software requirements	Implementation evidence	Effectiveness evidence
HA-01 Laser exposure	SYS-511 to 513	FW-502	DVP-CSI-502; SVR-CSI-001 (IEC 60825-1)	Accessible emission measurement with doors open and closed
HA-02 High voltage	SYS-522, 523, 532	FW-307, FW-601, FW-604 to 608, RA-1003	DVP-CSI-503; STS-CSI-010 §B	Fault injection: SI-10 hang, heartbeat loss, readback fault; HV removed in every trial
HA-04 Aerosol after clog	SYS-541, 543, 551 to 554	FW-210, FW-602, RA-504, RA-505	DVP-CSI-503, DVP-CSI-504	Aerosol challenge with simulated clog; no escape
HA-05 Access to sort chamber	SYS-521	FW-502	DVP-CSI-503	Door sequence test, 59 trials
HA-06 Moving parts	SYS-531 to 533, 562	FW-501, FW-601, FW-309	DVP-CSI-503, DVP-CSI-505	Pinch force measurement; e-stop timing
HA-07 Breakoff drift, low purity	SYS-542	FW-205, RA-501, RA-502	DVP-CSI-503; ITS-CSI-010	Induced breakoff drift; charging stops within 100 ms; purity maintained
HA-08 Wrong result (configuration, QC, calibration)	SYS-622, 623, 704, 903	FW-302, AD-102 to 105, AD-203, AD-903, RA-803, RA-805, RA-901	DVP-CSI-601, DVP-CSI-701; STS-CSI-010 §A, §C	Seeded errors (wrong filter, missing QC, overdue calibration, mismatched laser) are blocked
HA-09 Overpressure	SYS-524, 561	FW-401, FW-403	DVP-CSI-505	Relief valve test; depressurization within 10 s
HA-10 Chemical and decontamination	SYS-553	—	DVP-CSI-504	Material compatibility after 500 decontamination cycles
HA-11 Tip-over	SYS-563	—	DVP-CSI-505	10° tilt test
HA-12 Electrical shock and fire	SYS-314, 315, 501, 502	FW-308, FW-606	DVP-CSI-501; SVR-CSI-001 (IEC 61010)	Accredited laboratory results
HA-13 Electromagnetic interference	SYS-503	—	DVP-CSI-501; SVR-CSI-001 (IEC 61326)	Immunity tests with performance monitoring
HA-14 Fluid misconnection	SYS-801, 905	—	DVP-CSI-801, DVP-CSI-901	Misconnection attempts not possible
HA-15 Fluid run-out during sort	SYS-493	RA-207, FW-402	DVP-CSI-408	Low sheath and full waste scenarios blocked before sort

Table 1. Risk control verification. Use-related risk controls are also evaluated in the summative usability study (Part 2), which is reported separately.

9. Software and Subsystem Verification Inputs

System verification uses results from lower-level verification where the lower-level test is the best evidence for a system requirement. These results are referenced, not repeated.

Source	Plan and report	Used as evidence for
Software unit, integration and system test	SVP-CSI-010; UTS, ITS and STS-CSI-010; SVR-SW-CSI-010	SYS 6xx and 7xx; software parts of risk controls (Table 1)
Software architecture verification	SAD-SW-CSI-011 §16	Segregation and timing of safety functions
Subsystem design verification	Subsystem verification reports for SSD-CSI-100 to SSD-CSI-950	Inputs to SYS 4xx performance tests (not a substitute for them)
Test method validation	TMV-CSI-001	Validity of test methods used in DVP protocols
Compliance testing	SVR-CSI-001 (accredited external laboratory)	SYS 501 to 503, 511 to 513

Software release 3.0 must have passed software system test (STS-CSI-010) with no open anomalies of severity 1 or 2 before system verification of Sections 6 and 7 of the SRS is closed.

10. System Requirements Verification Matrix

Table 2 lists every system requirement in SRS-CSI-001 Rev D (146 requirements), the verification method stated in the SRS, and the protocol that verifies it. It is generated from the SRS so that no requirement is omitted. The acceptance criterion is the requirement statement unless Section 7 defines a statistical criterion.

ID	Requirement	Method	Protocol
SYS-301	The sorter shall not exceed 71 cm height, 122 cm width and 71 cm depth (28 × 48 × 28 in), excluding the workstation.	I	DVP-CSI-301
SYS-302	The sorter shall weigh no more than 185 kg (408 lb) in its heaviest configuration.	I	DVP-CSI-301
SYS-303	The fluidics supply module shall not exceed 66 cm height, 81 cm width and 66 cm depth (26 × 32 × 26 in), and shall weigh no more than 82 kg (180 lb) without tanks.	I	DVP-CSI-301
SYS-304	The fluidics supply module shall be mobile on lockable casters with leveling feet.	I	DVP-CSI-301
SYS-305	The sorter shall operate on a laboratory bench or on an optional bench no larger than 85.1 × 266.7 × 134.6 cm.	I	DVP-CSI-301
SYS-306	The umbilical between sorter and fluidics supply module shall allow separation of up to 2.7 m (9 ft).	I	DVP-CSI-301
SYS-311	The system shall operate from 100, 115 or 230 VAC (±10%), 50 or 60 Hz, single phase.	T	DVP-CSI-301
SYS-312	Total power consumption of the sorter and fluidics supply module shall not exceed 1,500 W.	T	DVP-CSI-301
SYS-313	The system shall require no more than one 20 A outlet for the sorter and fluidics supply module combined.	A	DVP-CSI-301
SYS-314	The fluidics supply module shall be powered and switched only from the sorter.	I	DVP-CSI-301
SYS-315	The system shall be protected by a resettable circuit breaker.	I	DVP-CSI-301
SYS-316	The system shall meet installation category II and pollution degree 2 as defined in IEC 61010-1.	A	DVP-CSI-301
SYS-321	The system shall meet all performance requirements at ambient temperatures of 15 to 30 °C.	T	DVP-CSI-301
SYS-322	The system shall meet all performance requirements at 5 to 80% relative humidity, non-condensing.	T	DVP-CSI-301
SYS-323	The system shall operate at altitudes up to 2,000 m.	A	DVP-CSI-301
SYS-324	The system shall withstand storage at 1 to 40 °C without loss of performance.	T	DVP-CSI-301
SYS-325	The system shall require no special room facilities, no external compressed air and no external vacuum.	D	DVP-CSI-301
SYS-326	The system shall accept optional facility air at 80 to 100 PSI, dry and oil-free.	T	DVP-CSI-301
SYS-331	Acoustic noise at the operator position shall not exceed 62 dBA during sorting.	T	DVP-CSI-301
SYS-401	The coefficient of variation (CV) of the full G0/G1 peak for propidium iodide-stained chicken erythrocyte nuclei, excited at 488 nm, shall be less than 3.0% (area parameter).	T	DVP-CSI-401
SYS-402	The CV of the full G0/G1 peak for Hoechst-stained chicken erythrocyte nuclei, excited at 405 nm or 375 nm, shall be less than 3.5% (area parameter).	T	DVP-CSI-401
SYS-411	The doublet-to-singlet ratio for chicken erythrocyte nuclei shall be 1.95 to 2.05, for PI excited at 488 nm and for Hoechst excited at 405 nm or 375 nm.	T	DVP-CSI-401
SYS-421	Forward and side scatter shall resolve 0.5 µm beads from instrument noise.	T	DVP-CSI-402
SYS-422	Scatter sensitivity shall allow separation of fixed platelets from noise and identification of bacteria.	T	DVP-CSI-402

ID	Requirement	Method	Protocol
SYS-423	The forward scatter path shall accept interchangeable neutral density filters of 0.5, 1.0, 1.5 and 2.0 for small and large particles.	I	DVP-CSI-402
SYS-431	The system shall support up to six installed lasers, with up to four active at the same time.	D	DVP-CSI-403
SYS-432	The 488 nm and 633 nm lasers shall be standard in every configuration. 375, 405, 445 and 561 nm lasers shall be available as options.	I	DVP-CSI-403
SYS-433	Minimum laser power out of the laser head shall be: 375 nm > 7 mW; 405 nm > 50 mW; 445 nm > 20 mW; 488 nm > 20 mW; 561 nm > 50 mW; 633 nm > 18 mW.	T	DVP-CSI-403
SYS-434	Elliptical beams at the interrogation point shall measure $9 \pm 3 \mu\text{m}$ in height and $65 \pm 7 \mu\text{m}$ in width.	T	DVP-CSI-403
SYS-435	Each active laser shall be focused on a spatially separated beam spot. Lasers sharing a spot (488/445 and 405/375) shall be mutually exclusive.	T	DVP-CSI-403
SYS-436	Optical alignment shall be fixed at the factory and shall require no user adjustment.	D	DVP-CSI-403
SYS-437	The system shall reach full performance within 30 minutes of laser power-on.	T	DVP-CSI-403
SYS-438	Lasers shall be air-cooled and shall need no external cooling or special power.	I	DVP-CSI-403
SYS-441	The system shall measure forward scatter, side scatter and at least 18 fluorescence parameters simultaneously in the fully populated configuration.	T	DVP-CSI-402
SYS-442	Each fluorescence detector shall have a user-replaceable bandpass filter and each detector array shall use user-replaceable longpass dichroic mirrors.	I	DVP-CSI-402
SYS-443	The system shall report height, area and width for every detected parameter.	T	DVP-CSI-402
SYS-444	Signals from each beam spot shall be time-aligned to the same event, with inter-laser delay set automatically during daily QC.	T	DVP-CSI-402
SYS-445	Data shall be digitized with a linear dynamic range of at least 18 bits.	A	DVP-CSI-402
SYS-451	The system shall acquire at least 70,000 events per second with 8 parameters and 12 compensation pairs without loss of data.	T	DVP-CSI-404
SYS-452	The system shall acquire more than 100,000 events per second when fewer parameters are recorded.	T	DVP-CSI-404
SYS-453	Sample flow rate shall be user-selectable over at least 10 to 80 $\mu\text{L}/\text{min}$.	T	DVP-CSI-404
SYS-461	For a four-way sort at 70 PSI, 90 kHz drop drive and an average threshold rate of 25,000 events per second, sort purity shall exceed 98% and yield shall exceed 80% of Poisson expected yield.	T	DVP-CSI-405
SYS-462	The system shall sort into up to four populations at the same time.	D	DVP-CSI-405
SYS-463	Drop drive frequency shall be adjustable from 1 to 100 kHz or wider.	T	DVP-CSI-405
SYS-464	The system shall resolve particle position within the drop to 1/32 of a drop and shall provide user-selectable purity, yield and single-cell sort modes.	T	DVP-CSI-405
SYS-465	The system shall automatically determine drop delay using fluorescent drop-delay beads.	T	DVP-CSI-405
SYS-466	Sorted cell lines and human PBMCs shall remain viable and proliferate after sorting at every default sheath pressure.	T	DVP-CSI-405
SYS-467	The system shall provide nozzles of 70, 85 and 100 μm , with a 130 μm nozzle as an option, each with a default sort setup.	I	DVP-CSI-405
SYS-468	Changing the nozzle shall not require optical realignment.	D	DVP-CSI-405
SYS-471	Sample carryover from one tube to the next shall not exceed 0.5% after the automatic sample line flush.	T	DVP-CSI-406

ID	Requirement	Method	Protocol
SYS-472	The system shall flush the inside and outside of the sample line with sheath after every tube and shall provide a sample line backflush on demand.	D	DVP-CSI-406
SYS-481	The system shall provide two-way sorting into 12 × 75 mm and 15 mL tubes, and four-way sorting into 12 × 75 mm tubes, 1.5 mL tubes and 1 mL microtubes.	D	DVP-CSI-407
SYS-482	With the plate deposition option, the system shall sort into 6-, 24-, 48-, 96- and 384-well plates and onto microscope slides.	D	DVP-CSI-407
SYS-483	With the plate deposition option, the system shall deposit single cells so that at least 95% of target wells receive exactly one particle.	T	DVP-CSI-407
SYS-484	The system shall record the measured parameters of each cell deposited in a plate well (index sorting).	D	DVP-CSI-407
SYS-485	The system shall allow users to define custom collection devices.	D	DVP-CSI-407
SYS-491	Sheath pressure shall be software-controlled from 5 to 75 PSI.	T	DVP-CSI-408
SYS-492	The fluidics supply module shall hold at least 10 L of sheath, 10 L of waste, 5 L of shutdown ethanol and three 5 L cleaning fluids.	I	DVP-CSI-408
SYS-493	The system shall warn the user before sheath runs out or waste is full, and shall prevent sorting when either condition would interrupt a sort.	T	DVP-CSI-408
SYS-494	The sample chamber shall accept 15 mL, 12 × 75 mm and 1 mL tubes.	I	DVP-CSI-408
SYS-495	The system shall agitate the sample tube at user-selectable speeds.	D	DVP-CSI-408
SYS-4101	The sample chamber shall hold a user-selected temperature, reached within 45 minutes with the chamber closed.	T	DVP-CSI-408
SYS-4102	With the temperature control option, collection tubes shall be held at a user-selected temperature within ±2 °C.	T	DVP-CSI-408
SYS-501	The system shall comply with IEC 61010-1 and IEC 61010-2-101 for protection against electric shock and fire.	T	DVP-CSI-501
SYS-502	Protective earth continuity, dielectric strength and touch current shall be tested on every unit in production.	T	DVP-CSI-501
SYS-503	The system shall comply with IEC 61326-1 and IEC 61326-2-6 for electromagnetic compatibility.	T	DVP-CSI-501
SYS-511	The system shall be a Class 1 laser product per IEC 60825-1 in all user-accessible conditions.	T	DVP-CSI-502
SYS-512	Opening the flow cell access door shall block all laser beams by a shutter before any beam is accessible.	T	DVP-CSI-502
SYS-513	Laser warning labels shall be placed on all service-access panels behind which Class 3B radiation is present.	I	DVP-CSI-502
SYS-521	The sort collection chamber door shall not open unless the sort block door is closed.	T	DVP-CSI-503
SYS-522	An illuminated warning light and a software indicator shall show whenever deflection plate voltage is on.	T	DVP-CSI-503
SYS-523	Deflection plate voltage shall not exceed 12,000 V between plates and shall be insulated so that no energized part is accessible with the sort block door closed.	T	DVP-CSI-503
SYS-524	The system shall fully depressurize within 10 seconds of main power switch-off.	T	DVP-CSI-503
SYS-531	The system shall provide an illuminated emergency stop button next to the sample loading port.	I	DVP-CSI-503
SYS-532	Pressing the emergency stop shall, within 2 seconds: unload the sample tube, stop plate stage motion, turn off the stream, turn off deflection plate voltage, close the waste aspirator drawer and display a warning.	T	DVP-CSI-503
SYS-533	The emergency stop button shall light while the sample loading port is moving.	T	DVP-CSI-503

ID	Requirement	Method	Protocol
SYS-541	The system shall monitor the drop breakoff continuously during sorting.	T	DVP-CSI-503
SYS-542	If the breakoff moves outside its tolerance, the system shall pause sorting until the breakoff recovers, while sample continues to flow.	T	DVP-CSI-503
SYS-543	If a nozzle clog is detected, the system shall, within 2 seconds: stop the stream, turn off plate voltage, close the aspirator drawer and unload the sample tube.	T	DVP-CSI-503
SYS-551	With the sort block and collection chamber doors closed and the aerosol evacuation option running, no aerosolized particles shall be detected outside the sort chamber during sorting or during a simulated clog.	T	DVP-CSI-504
SYS-552	The aerosol evacuation option shall provide at least 7 CFM in normal mode and at least 30 CFM in boost mode through a ULPA filter with efficiency above 99.9995% at 0.12 μ m.	T	DVP-CSI-504
SYS-553	All wetted surfaces and surfaces exposed to specimens shall withstand the decontamination agents specified in the IFU (10% bleach, 70% ethanol).	T	DVP-CSI-504
SYS-554	The IFU shall describe safe handling of a clog and decontamination after sorting infectious specimens.	I	DVP-CSI-504
SYS-561	Pressurized tanks shall be rated for at least 1.5 times the maximum operating pressure and shall include overpressure relief.	A, T	DVP-CSI-505
SYS-562	Moving parts (loading port, plate stage, aspirator drawer) shall not exert a pinching force greater than the limits of IEC 61010-1.	T	DVP-CSI-505
SYS-563	The system shall remain stable when tilted 10° in any direction.	T	DVP-CSI-505
SYS-601	The system shall perform an automated fluidics startup that primes all fluid lines and removes air.	D	DVP-CSI-601
SYS-602	The system shall perform an automated fluidics shutdown that leaves the flow path filled with 70% ethanol.	D	DVP-CSI-601
SYS-603	The system shall provide automated cleaning modes: flow cell clean, sample line backflush, prime after tank refill, and a bleach, water and ethanol clean cycle.	D	DVP-CSI-601
SYS-604	The system shall be ready to run daily QC within 45 minutes of power-on, including laser warm-up.	T	DVP-CSI-601
SYS-611	Selecting a nozzle size in software shall load a verified default setup for sheath pressure, drop drive, breakoff targets, drop delay, plate voltage, stream charges and laser delays.	D	DVP-CSI-601
SYS-612	The system shall display a live image of the drop breakoff and of the side streams.	D	DVP-CSI-601
SYS-613	The system shall hold the breakoff position automatically by adjusting drop drive amplitude.	T	DVP-CSI-601
SYS-614	The system shall provide a test sort function that generates side streams for adjustment without sample.	D	DVP-CSI-601
SYS-621	The system shall run a daily bead-based performance check that measures detector performance, sets inter-laser delay and records results with pass/fail status.	T	DVP-CSI-601
SYS-622	The system shall not allow an IVD assay to run unless the most recent performance check for the active configuration has passed within the last 24 hours.	T	DVP-CSI-601
SYS-623	The software configuration shall identify the installed lasers, mirrors and filters, and shall warn the user if the selected configuration does not match the active lasers.	T	DVP-CSI-601
SYS-624	Performance check results shall be stored and trended so users can see changes over time.	D	DVP-CSI-601
SYS-631	The system shall allow users to define gates, sort populations and sort layouts for each collection device.	D	DVP-CSI-601
SYS-632	The system shall calculate fluorescence compensation from single-stained controls.	T	DVP-CSI-601

ID	Requirement	Method	Protocol
SYS-633	The system shall show live sort counters for each sort population and the sort efficiency.	D	DVP-CSI-601
SYS-634	The user shall be able to pause, resume and stop a sort without losing the sort record.	D	DVP-CSI-601
SYS-635	The system shall produce a sort report giving the sort settings, counts and events for each sort.	D	DVP-CSI-601
SYS-701	The software shall require a unique user login and support at least operator, supervisor and administrator roles with different permissions.	T	DVP-CSI-701
SYS-702	The software shall keep a secure, computer-generated, time-stamped audit trail of record creation, modification and deletion that users cannot edit, as required by 21 CFR Part 11.	T	DVP-CSI-701
SYS-703	The software shall support electronic signatures for result release and assay approval.	T	DVP-CSI-701
SYS-704	Approved IVD assay templates shall be locked so that operators cannot change instrument settings, gates or analysis once approved.	T	DVP-CSI-701
SYS-705	The software shall export data in FCS 3.1 format and reports in PDF.	T	DVP-CSI-701
SYS-706	The software shall detect and report loss of communication with the instrument without corrupting stored data.	T	DVP-CSI-701
SYS-707	The workstation shall have a locked, validated operating system image, with only the services needed for operation enabled.	I	DVP-CSI-701
SYS-708	The manufacturer shall maintain a software bill of materials and a process for assessing and patching vulnerabilities.	I	DVP-CSI-701
SYS-709	User data at rest shall be protected by access control, and data exported off the workstation shall be integrity-protected.	T	DVP-CSI-701
SYS-801	The sorter and fluidics supply module shall connect through a keyed umbilical carrying power, sheath, waste, air and a serial communication line; connectors shall not be interchangeable.	I	DVP-CSI-801
SYS-802	The sorter shall connect to the workstation by Ethernet on a dedicated, isolated network link.	I	DVP-CSI-801
SYS-803	The system shall provide connections for the aerosol evacuation module and the temperature control option without tools.	D	DVP-CSI-801
SYS-804	Collection tube holders shall share a universal top with evacuation ports compatible with the aerosol evacuation option.	I	DVP-CSI-801
SYS-805	The system shall use sheath fluid of buffered saline or 1× PBS, and cleaning fluids of 10% bleach, deionized water and 70% ethanol.	T	DVP-CSI-801
SYS-806	The system shall support IVD reagents and QC, alignment and drop-delay beads as specified in ACC-CSI-001.	T	DVP-CSI-801
SYS-901	A trained operator shall be able to complete startup, daily QC, a four-way sort and shutdown using only the IFU and on-screen guidance.	D	DVP-CSI-901
SYS-902	All critical tasks identified in the use-related risk analysis shall be completed without use errors that could cause harm in summative usability testing.	T	DVP-CSI-901
SYS-903	Tube holders, nozzles and collection devices shall be marked with their size and keyed or shaped to prevent incorrect installation.	I	DVP-CSI-901
SYS-904	The software shall warn the user before any action that would discard unsaved data or stop an active sort.	D	DVP-CSI-901
SYS-905	Fluid containers shall be labeled and color-coded by content, and their connectors keyed to prevent misconnection.	I	DVP-CSI-901

ID	Requirement	Method	Protocol
SYS-1001	The system shall have a design service life of at least 7 years with scheduled maintenance.	A	DVP-CSI-1001
SYS-1002	Mean time between failures requiring a service visit shall be at least 2,000 operating hours.	A, T	DVP-CSI-1001
SYS-1003	Routine user maintenance (filters, sample lines, waste caps, nozzle cleaning) shall be possible without tools other than those in the accessory kit, and shall not need realignment.	D	DVP-CSI-1001
SYS-1004	User maintenance intervals shall be no more frequent than weekly for filter purging and every 3 months for sheath filter replacement.	A	DVP-CSI-1001
SYS-1005	Lasers, PMTs, detector arrays, PCBAs and the fluidics supply module shall be field-replaceable by a service engineer.	D	DVP-CSI-1001
SYS-1006	A trained service engineer shall be able to replace any field-replaceable unit and return the system to passing QC within 4 hours.	D	DVP-CSI-1001
SYS-1007	The system shall provide diagnostic data (pressures, laser status, voltages, error logs) to service engineers without access to patient data.	D	DVP-CSI-1001
SYS-1008	Lasers and detector arrays shall be upgradable in the field according to CFG-CSI-001.	D	DVP-CSI-1001
SYS-1101	Product labels shall meet ISO 15223-1 and ISO 18113-3 and shall include the UDI in human-readable and machine-readable form.	I	DVP-CSI-1101
SYS-1102	The IFU (Operator's Guide) shall include intended use, limitations, warnings, installation requirements, operation, QC, maintenance and troubleshooting.	I	DVP-CSI-1101
SYS-1103	Laser, high-voltage, biohazard and pinch-point labels shall be placed where the hazard exists, as defined in LBL-CSI-001.	I	DVP-CSI-1101
SYS-1104	The system shall be supplied with a site preparation guide and a service manual.	I	DVP-CSI-1101
SYS-1105	Labeling and IFU shall be available in the languages required by each market.	I	DVP-CSI-1101
SYS-1201	The shipping configuration shall protect the sorter and the fluidics supply module so that, after ISTA 3E transport testing, the system passes installation qualification without optical realignment.	T	DVP-CSI-1201
SYS-1202	Shipping crates shall carry shock and tilt indicators.	I	DVP-CSI-1201
SYS-1203	Installation, including installation qualification, shall be completed by a service engineer in one working day.	D	DVP-CSI-1201
SYS-1204	The system shall pass through a standard 90 cm (36 in) laboratory door with covers or crate removed.	I	DVP-CSI-1201
SYS-1301	The system shall be designed and manufactured under a quality management system that meets 21 CFR 820 (QMSR) and ISO 13485:2016.	I	DVP-CSI-1301
SYS-1302	The system shall meet the requirements for US marketing as a Class II IVD instrument through premarket notification (510(k)).	A	DVP-CSI-1301
SYS-1303	The system shall meet the general safety and performance requirements of EU IVDR 2017/746 (Annex I).	A	DVP-CSI-1301
SYS-1304	Risk management shall be carried out according to ISO 14971:2019 and all residual risks shall be acceptable under RMP-CSI-001.	A	DVP-CSI-1301
SYS-1305	Software shall be developed according to IEC 62304 and IEC 82304-1.	I	DVP-CSI-1301
SYS-1306	The system shall comply with applicable restrictions on hazardous substances (EU RoHS) and with WEEE marking requirements.	I	DVP-CSI-1301

Table 2. System requirements verification matrix.

PART 2. DESIGN VALIDATION

Design validation confirms that the finished CSI-100 meets the needs of its users and its intended use. It is traced to the user needs (UN-nn), not to system requirements, and is performed with production-equivalent units by intended users under actual or simulated use conditions.

11. Validation Approach

11.1 Intended use under validation

The CSI-100 is an automated, multi-laser flow cytometer and droplet cell sorter for identification, enumeration and physical separation of cell populations in stained human specimens, for in vitro diagnostic use by trained laboratory professionals in clinical laboratories. Validation covers the instrument functions that support this intended use; each IVD assay is validated clinically under its own plan.

11.2 Conditions

- Units: VAL-U1 to VAL-U4 from the validation build, made with released DMR documents and processes (DTP-CSI-001).
- Software: the release candidate intended for market release, installed by service engineers using the released installation procedure.
- Labeling: final Operator's Guide, labels and training materials.
- Environment: three external clinical laboratories and one internal simulated-use laboratory.
- Users: clinical laboratory scientists trained with the commercial training program, not members of the design team.
- Specimens: de-identified residual clinical specimens and contrived samples where specimens are unavailable.

11.3 Entry criteria

Entry criterion	Evidence
System requirements relevant to each study are verified with no open critical anomalies	PVR-CSI-001 and SVR-CSI-001 sections; trace matrix status
Validation units built and released from the PQ build	DHRs for VAL-U1 to VAL-U4
Software release candidate passed software system test	SVR-SW-CSI-010
Final labeling and training materials approved	LBL-CSI-001, OG-CSI-001 Rev B
Ethics approval or exemption for specimen use at each site	Site approvals on file
Validation protocols approved	VAL-CSI-01 to VAL-CSI-12 in DVAL-CSI-001

12. Validation Studies and Acceptance Criteria

Study	Title	User needs	Design	Acceptance criteria
VAL-CSI-01	Method comparison	UN-01, UN-03	Stained clinical specimens run on the CSI-100 and on the reference flow cytometer; percentage of major lymphocyte and CD34+ populations compared.	Deming regression slope 0.90 to 1.10 and $r \geq 0.95$ for each population; bias within ± 2 percentage points or $\pm 10\%$ (whichever is larger).
VAL-CSI-02	Clinical sort purity, recovery and viability	UN-02, UN-05, UN-15	Target populations sorted from clinical specimens with each nozzle; sorted fractions re-analyzed; viability by dye exclusion.	Purity $\geq 95\%$ (re-analysis), recovery $\geq 70\%$ of expected, viability $\geq 85\%$, for every specimen type.
VAL-CSI-03	Summative usability (human factors validation)	UN-06, UN-09	15 intended users per user group perform critical tasks (startup, QC, sort setup, clog recovery, decontamination, shutdown) with final labeling.	No use errors on critical tasks that could lead to harm; residual use-related risks acceptable per RMR-CSI-001.
VAL-CSI-04	Multi-site precision	UN-08, UN-01	3 sites $\times$ 5 days $\times$ 2 runs $\times$ 2 replicates of stabilized control material.	Total CV $\leq 10\%$ for populations $> 10\%$ of events; $\leq 15\%$ for populations 1% to 10%; QC passes on every day.
VAL-CSI-05	Laboratory workflow and throughput	UN-04	Sites run a typical day of specimens, including QC, acquisition, sorting and review.	Median time from power-on to first result ≤ 60 min; laboratory scientists rate workflow acceptable ($\geq 80\%$ agree).
VAL-CSI-06	Installation in clinical laboratories	UN-07	Service engineers install units at each site using the site preparation guide.	Installation completed in one day with no facility changes beyond the site preparation guide.
VAL-CSI-07	Carryover with clinical specimens	UN-14	High-count specimens followed by low-count or negative specimens, per site.	No clinically meaningful carryover: low-count results within precision limits of VAL-CSI-04.
VAL-CSI-08	Temperature control with cells	UN-11	Temperature-sensitive cell samples held and sorted at 4 °C and 37 °C.	Sample and collection temperatures within user settings; viability $\geq 85\%$.
VAL-CSI-09	Records, review and release in use	UN-10	Supervisors review and release results, export to the site LIS, audit results in the Data Management Application.	All released results traceable to user, instrument, QC and template version; laboratory quality staff confirm Part 11 controls fit their procedures.
VAL-CSI-10	Serviceability in the field	UN-12	Scheduled maintenance by users and one unscheduled repair by a service engineer at each site.	User maintenance completed with the Operator's Guide; repair and return to passing QC in ≤ 4 h.
VAL-CSI-11	Configuration and field upgrade	UN-16	Field upgrade of a laser and detector array on one unit; laboratory runs its panels after upgrade.	Upgrade completed per service manual; post-upgrade QC and panel results meet VAL-CSI-01 criteria.
VAL-CSI-12	Regulatory and labeling review	UN-13	Review of the complete device, labeling and IFU against US and EU requirements by Regulatory Affairs.	No gaps against 21 CFR 809.10 and IVDR Annex I and Annex I Chapter III labeling requirements.

13. Sites, Participants and Specimens

Site	Type	Units	Studies
Site 1	Hospital clinical flow cytometry laboratory	VAL-U1 (4-laser, all options)	01, 02, 03, 04, 05, 06, 07, 08, 09, 10
Site 2	Reference laboratory	VAL-U2 (6-laser)	01, 02, 04, 05, 06, 07, 09, 11
Site 3	Cell therapy processing laboratory	VAL-U3 (4-laser)	02, 04, 05, 06, 07, 10
Internal	Simulated-use laboratory	VAL-U4 (2-laser)	03 (usability), 12

Participants in VAL-CSI-03 are recruited to represent the intended user groups (operators and supervisors), with at least 15 participants per group, and receive the same training a customer would receive. Specimens are de-identified residual samples collected under each site's approved procedures; the number of specimens per study is set in each protocol from the precision needed for its acceptance criteria (for example, at least 100 specimens for VAL-CSI-01).

COMMON

14. Traceability

Two separate trace tables are maintained: one for verification (user need to system requirements to protocols) and one for validation (user need to validation studies). Both are generated from SRS-CSI-001 and this plan and are exported to TM-CSI-DT-01.

14.1 Verification trace: user needs to system requirements to protocols

User need	System requirements (verified in Part 1)	Protocols
UN-01	SYS-401, SYS-402, SYS-411, SYS-421, SYS-422, SYS-433, SYS-434, SYS-443, SYS-444, SYS-445, SYS-495, SYS-632, SYS-704, SYS-806	401, 402, 403, 408, 601, 701, 801
UN-02	SYS-461, SYS-462, SYS-464, SYS-466, SYS-631, SYS-633, SYS-634	405, 601
UN-03	SYS-431, SYS-432, SYS-435, SYS-441, SYS-632	402, 403, 601
UN-04	SYS-451, SYS-452, SYS-453, SYS-492, SYS-494, SYS-604	404, 408, 601
UN-05	SYS-464, SYS-481, SYS-482, SYS-483, SYS-484, SYS-485, SYS-631	405, 407, 601
UN-06	SYS-551, SYS-803, SYS-804, SYS-902, SYS-1304	504, 801, 901, 1301
UN-07	SYS-301, SYS-302, SYS-303, SYS-304, SYS-305, SYS-306, SYS-311, SYS-312, SYS-313, SYS-321, SYS-322, SYS-324, SYS-325, SYS-326, SYS-331, SYS-438, SYS-1104, SYS-1201, SYS-1203, SYS-1204	301, 403, 1101, 1201
UN-08	SYS-436, SYS-437, SYS-444, SYS-465, SYS-468, SYS-613, SYS-621, SYS-622, SYS-624, SYS-806, SYS-1201	402, 403, 405, 601, 801, 1201
UN-09	SYS-437, SYS-465, SYS-492, SYS-493, SYS-601, SYS-602, SYS-603, SYS-611, SYS-612, SYS-614, SYS-805, SYS-901	403, 405, 408, 601, 801, 901
UN-10	SYS-635, SYS-701, SYS-702, SYS-703, SYS-705, SYS-706, SYS-709, SYS-802, SYS-904, SYS-1007	601, 701, 801, 901, 1001
UN-11	SYS-4101, SYS-4102, SYS-803	408, 801
UN-12	SYS-1001, SYS-1002, SYS-1003, SYS-1004, SYS-1005, SYS-1006, SYS-1007, SYS-1104, SYS-1202	1001, 1101, 1201
UN-13	SYS-1101, SYS-1102, SYS-1105, SYS-1301, SYS-1302, SYS-1303, SYS-1304, SYS-1305, SYS-1306	1101, 1301
UN-14	SYS-471, SYS-472, SYS-603	406, 601
UN-15	SYS-423, SYS-463, SYS-467, SYS-491, SYS-611	402, 405, 408, 601
UN-16	SYS-431, SYS-432, SYS-442, SYS-623, SYS-1008	402, 403, 601, 1001

Protocol numbers shown without the DVP-CSI- prefix. System requirements whose source is a hazard or standard only are traced in Table 1 and Table 2.

14.2 Validation trace: user needs to validation studies

User need	Statement	Validation studies (Part 2)
UN-01	Identify and enumerate cell populations in stained human specimens with results accurate and precise enough for clinical use.	VAL-CSI-01, VAL-CSI-04
UN-02	Physically separate target cell populations at high purity and recovery, with cells remaining viable for downstream analysis.	VAL-CSI-02
UN-03	Measure many markers on each cell at the same time (multicolor panels).	VAL-CSI-01
UN-04	Process specimens at a throughput that fits the laboratory workday.	VAL-CSI-05
UN-05	Collect sorted cells into tubes, multiwell plates and slides, including single-cell deposition with a record of each deposited cell.	VAL-CSI-02
UN-06	Operate the instrument without risk of injury from laser light, high voltage, moving parts, pressure or biohazardous aerosols.	VAL-CSI-03
UN-07	Install the instrument in a standard clinical laboratory without special facilities.	VAL-CSI-06
UN-08	Obtain consistent performance from day to day with minimal user adjustment, confirmed by a daily quality control check.	VAL-CSI-04
UN-09	Set up, run, clean and shut down the instrument through guided, largely automated procedures.	VAL-CSI-03
UN-10	Keep results and records secure, attributable and traceable, with controlled user access.	VAL-CSI-09
UN-11	Keep samples and sorted cells at a chosen temperature during the run.	VAL-CSI-08
UN-12	Keep the instrument in service with predictable maintenance and fast repair.	VAL-CSI-10
UN-13	Use the instrument in the United States and the European Union in compliance with IVD regulations.	VAL-CSI-12
UN-14	Avoid carryover and cross-contamination between specimens.	VAL-CSI-07
UN-15	Handle cells of different sizes and fragility.	VAL-CSI-02
UN-16	Choose the laser and detector configuration that fits the laboratory's assays, and upgrade it later.	VAL-CSI-11

Coverage check: 16 user needs; 0 without a validation study; 0 without a verified system requirement. 146 system requirements, each assigned a protocol in Table 2.

15. Anomalies, Deviations and Re-Verification

15.1 Anomalies

Severity	Definition	Required action
1 Critical	Safety risk or failure of a risk control	Stop testing in the affected area; root cause; design change; re-verify
2 Major	Requirement not met	Root cause; correction; re-verify before closure
3 Minor	Requirement met but unexpected behavior	Evaluate; correct or justify
4 Cosmetic	No effect on function or safety	Track

Anomalies are recorded in the defect tracking system and reviewed weekly by the anomaly review board. A test failure caused by the test setup or test method is documented and re-run only after the cause is shown; test failures are never discarded.

15.2 Deviations

Any departure from an approved protocol is recorded as a deviation with its justification and impact, and approved by Quality before the affected result is used.

15.3 Re-verification and regression after change

Design changes after verification are assessed through the Engineering Change process. The assessment identifies the affected system requirements, risk controls and user needs using the trace tables, and defines the re-verification and, where user needs are affected, re-validation required. Software changes follow the regression rules in SVP-CSI-010.

16. Deliverables

Deliverable	Content	Owner
DVP-CSI-301 to DVP-CSI-1301	Approved verification protocols	V&V Engineering
PVR-CSI-001	System Performance Verification Report (SRS Sections 3, 4, 6, 8 to 12)	V&V Engineering
SVR-CSI-001	Safety, EMC and Laser Compliance Test Report (SRS Section 5.1, 5.2)	Compliance
SVR-SW-CSI-010	Software Verification Report (referenced)	Software Engineering
DVAL-CSI-001	Design Validation Plan annex (VAL-CSI-01 to 12) and Design Validation Report	Clinical Affairs
HFE-CSI-001	Human factors engineering report, including VAL-CSI-03	Human Factors
TM-CSI-DT-01	Trace matrix with verification and validation status	Systems Engineering
VVR-CSI-001	V&V Summary Report: Part 1 verification, Part 2 validation, open items	Systems Engineering

17. Schedule

The schedule is aligned with the design transfer phases in DTP-CSI-001: verification uses pilot build units (T1); validation uses validation build units (T2) and starts after DTRR-2.

Activity	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun
V&V plan and protocols approved										
Physical, performance verification (DVP 301 to 408)										
Safety, EMC and laser testing (external)										
Safety function and containment (DVP 503, 504)										
Software system test (STS-CSI-010)										
Functional, software, interface (DVP 601 to 801)										
Reliability life test (DVP 1001)										
Transport and labeling (DVP 1101, 1201)										
Verification report PVR / SVR										
Validation build and site readiness (DTRR-2)										
Validation studies VAL-CSI-01 to 12										
Validation report DVAL-CSI-001										
V&V summary report VVR-CSI-001										
Regulatory submission										

Months: September 2026 to June 2027. Shaded cells show planned activity.

18. Reference Documents

18.1 External standards and regulations

Reference	Title
21 CFR Part 820 (QMSR)	Quality Management System Regulation (design verification and validation)
ISO 13485:2016 §7.3.6, §7.3.7	Design and development verification and validation
ISO 14971:2019	Application of risk management to medical devices
IEC 62304 / IEC 62366-1	Software life cycle processes / usability engineering
IEC 61010-1, 61010-2-101, 61326-1, 61326-2-6, 60825-1	Safety, EMC and laser safety
FDA guidance: Applying Human Factors and Usability Engineering to Medical Devices	Summative usability evaluation
CLSI EP05, EP09	Precision; method comparison and bias estimation
CLSI H62	Validation of assays performed by flow cytometry
ISTA 3E	Transport testing
21 CFR 809.10; EU IVDR 2017/746	IVD labeling; IVD regulation

18.2 Program documents

Document no.	Rev	Title
SRS-CSI-001	D	System Requirements Specification (design inputs; user needs)
SRS-SW-CSI-010	E	Software Requirements Specification: Research, Firmware, Admin and Data Management
SAD-SW-CSI-011	C	Software Architecture Description
SVP-CSI-010	B	Software Verification Plan
STS-CSI-010 / ITS-CSI-010	A	Software System / Integration Test Specifications
HA-CSI-001	C	Hazard Analysis
RMP-CSI-001 / RMR-CSI-001	B / A	Risk Management Plan / Report
TMV-CSI-001	A	Test Method Validation Protocols and Reports
PVR-CSI-001	A	System Performance Verification Report
SVR-CSI-001	A	Safety, EMC and Laser Compliance Test Report
DVAL-CSI-001	A	Design Validation Plan annex and Report
CFG-CSI-001	C	Configuration Definition: Laser and Detector Builds
DTP-CSI-001	B	Design Transfer Plan
TM-CSI-DT-01	B	Trace Matrix
OG-CSI-001 / LBL-CSI-001	B / B	Operator's Guide / Labeling Specification

Appendix A. Terms and Abbreviations

Term	Meaning
Design verification	Confirmation by objective evidence that design outputs meet design input requirements
Design validation	Confirmation by objective evidence that the device meets user needs and intended use
DV unit / VAL-U unit	Design verification unit (pilot build) / validation unit (validation build)
DVP / VAL	Design verification protocol / validation study protocol
DTRR	Design Transfer Readiness Review (DTP-CSI-001)
HA	Hazard in HA-CSI-001
Tolerance interval	Interval expected to contain a stated proportion of the population with a stated confidence
UN / SYS	User need / system requirement (SRS-CSI-001)